Employee Attrition: Key Drivers

Executive Summary

Prepared from a statistical analysis of 1,470 employee records, combining hypothesis testing with Bayesian modeling to identify which factors are most reliably linked to attrition, and how strong each effect is.

What we found

- **Overtime is the single biggest signal.** Employees who work overtime leave at roughly 4.5x the rate of those who don't — the clearest and strongest pattern in the data.
- **Tenure protects against attrition.** The longer an employee has worked, the less likely they are to leave — each additional step up in experience roughly halves the odds of attrition. Newer employees are consistently the highest-risk group.
- **Single employees leave more than divorced or married employees.** Single employees have about 3.4x the odds of leaving compared to divorced employees. Married and divorced employees show no meaningfully different attrition risk from each other.
- **Two smaller but real effects:** Employees living farther from work, and employees reporting lower job satisfaction, both show modestly higher attrition risk — real, but far smaller than overtime or tenure.

What this means

Overtime and tenure together tell a coherent story: newer employees under heavy workload are the group most likely to leave. If retention efforts need to be prioritized, this is the segment to start with — for example, reviewing overtime allocation for employees in their first few years, or building in extra support and check-ins during that period.

A caveat worth stating plainly

This analysis shows association, not proof of cause. We can't say from this data alone whether overtime causes people to leave, or whether people already planning to leave simply disengage and get assigned different workloads. The findings tell us where to look closer and where to prioritize — not a guaranteed fix.

Recommendation

Start with a focused look at overtime policy for employees with under ~2 years of tenure, since that's where the two strongest risk factors overlap. A small pilot — capping overtime or adding proactive manager check-ins for that group — would let the effect be tested directly rather than inferred statistically.

Methodology: hypothesis testing (chi-square, Welch's t-test, Mann-Whitney U) identified candidate drivers; a Bayesian logistic regression (PyMC, validated against a hand-implemented Metropolis-Hastings sampler) then estimated each driver's effect with full uncertainty quantification, reported as odds ratios with 94% credible intervals.